

Continuing with N1 Grid Engine

Job Submission and Scheduling at ACEnet
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Topics Covered

- advanced topics:
 - mail notification
 - altering jobs on the fly
 - special purpose queues
- troubleshooting N1GE jobs ☐
 - my job won't start!
 - useful utilities - qsum, showq, qghost
 - qacct - more info on failed jobs

Mail Notification

- ☐ notification on job success/failure
- directives:
 - specify email address:

`#$ -M michelles@mun.ca`

- specify when to notify:

`#$ -m eas` (end, abort, suspend)

Altering Jobs on the Fly

- two options - qdel, qalter
- qdel - delete job, modify script, then resubmit
- qalter - modify queued jobs:

qalter <modified job parameters> <job ID>

e.g.

qalter -l h_rt=48:0:0 12345

Special Purpose Queues

- test.q - test jobs, max. 1 hour

`qsub -l test=true myjob.sh`

- sub.q - subordinate queue
- scavenge extra CPU cycles from green ACEnet

`qsub -l susp=true myjob.sh`

Special Purpose Queues

- gaussian.q
- used for submitting Gaussian jobs
- can use -q gaussian.q for serial/parallel jobs or -pe gaussian 2-4 for parallel jobs

`qsub -q gaussian.q myg09job.sh`

- other queues for green ACEnet - demirov.q, cmms.q, tarasov.q and ivan.q

Troubleshooting N1GE jobs

- my job won't start:

- qsub refuses:

- > qsub nosub

- Unable to run job: error: no suitable queues.
Exiting.

- job in error state ("Eqw"):

- use qstat -j <job-ID>, look for "error reason"
 - usually due to incorrect paths

Troubleshooting N1GE Jobs

- not enough memory:
 - check memory left on nodes
 - remember: $\text{RAM requested} = h_v\text{mem} * \text{nslots!}$
- job remains in "qw" state:
 - priority is too low
 - not enough free slots
 - not enough slots in the right queue

Troubleshooting N1GE Jobs

- shared memory job slots:
 - remember - all slots must be on the same hosts
 - check host list for number of free and total slots
- parallel job starvation:
 - jobs schedule when resources become available
 - scheduler tries to fill available resources
 - small jobs starve out large jobs if the small jobs keep comping

Troubleshooting N1GE Jobs

- solution to parallel job starvation - reservation

`qsub -R y mylargejob.sh`

- as resources are freed, they are reserved for the large job
- reserved for highest priority jobs
- expensive, so use sparingly!

qsum

Queue Info Processors

Queue Name	IN	USE	AVAIL	UNAVAIL	TOTAL
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long.q	46	198	0	244	
medium.q	420	508	0	928	
short.q	587	721	0	1308	

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Users Info	Running	Waiting	Error/Other
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ID	#JOBS	#CPUS	#JOBS	#CPUS	#JOBS	#CPUS
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bboates	30	480	6	96	127	2032
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